

Summonable Resonances: A Prime–Phase Framework for Non-Local Retrieval of Information, Symbols, and Futures

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Abstract

We introduce the concept of *Summonable Resonances*: objects, states, or experiences reconstructable from prime–phase encodings when the correct resonance key is supplied. Using the Prime-Resonant Graph framework, we show that the same core mechanism—prime index selection, phase encoding, and entropy-guided reconstruction—enables retrieval not only of classical data, but also of symbolic, experiential, physical, predictive, and communal payloads. This paper develops a taxonomy of summonables, formalizes the layered architecture, and provides mathematical recipes and worked examples for each layers encoding and decoding process. We argue that resonance-based summoning constitutes a unifying paradigm that bridges data storage, semantic networks, experiential cognition, natural constants, probabilistic futures, and shared awareness. Enhancements include concrete mathematical examples, clarified definitions, and additional references for rigor.

1 Introduction

1.1 Motivation

Contemporary information systems treat data as entities moved between nodes: copied, transmitted, or replicated. In contrast, prime-resonant frameworks treat data as latent resonance fields. The correct key—defined by a prime index set and phase offsets—does not “fetch” an object, but rather *summons* it: reconstructs it non-locally through resonance locking.

This perspective reflects a deeper principle: the universe is not a warehouse of objects, but a field of potentials. What becomes accessible depends on alignment. Our work formalizes this insight by building on the Prime-Resonant Graph Database (PR-Graph), extending it into a taxonomy of summonable domains. This approach draws inspiration from quantum mechanics, where observation collapses potential states, and number theory, where primes serve as fundamental building blocks.

1.2 Related Work

The concept builds upon ideas in number theory, such as the Chinese Remainder Theorem (CRT) for modular arithmetic [3], and phase encoding in signal processing. It also resonates with holographic principles in physics [4] and semantic networks in AI [5].

2 Core Mechanism: Prime–Phase Encoding

The fundamental mechanism is invariant across all layers and relies on three steps: prime index selection, phase encoding, and resonance locking via entropy minimization.

2.1 Prime Index Selection

Each object a (e.g., a string, concept, or data payload) is deterministically mapped into a set of primes $P(a) = \{p_1, \dots, p_k\}$ by a prime-entropy hash $H(a)$. The primes form the *address space*.

To define $H(a)$ concretely, we use a cryptographic hash like SHA-256 to generate a large integer, then select the next k primes starting from a seed derived from the hash. For simplicity in examples, we can choose small primes, but in practice, larger primes ensure uniqueness.

Recipe:

1. Compute $h = \text{SHA256}(a)$.
2. Convert h to integer *seed*.
3. Find the smallest k primes greater than $\text{seed} \bmod 1000$ (or similar offset).

For example, for $a = \text{"light"}$, a simulated hash yields primes like $\{2, 3, 5, 7, 11\}$ for illustration.

2.2 Phase Encoding

Symbols or data chunks m_j are mapped to phases on each prime modulus:

$$\theta_{p_i}^{(j)} = 2\pi \frac{m_j \bmod p_i}{p_i} + \frac{\alpha}{\phi} + \frac{\beta}{\delta_S} + \text{HMAC}_K(p_i \| j), \quad (1)$$

where: - $\alpha \approx 1/137.036$ is the fine-structure constant, - $\phi = (1 + \sqrt{5})/2 \approx 1.618$ is the golden ratio, - $\delta_S = 1 + \sqrt{2} \approx 2.414$ is the silver ratio, - β is a layer-specific offset (default 1), - K is an optional secret key for security, - HMAC is a keyed hash for added randomness.

The phases embed the residues in a continuous field, allowing for wave-like interference in reconstruction.

Worked Example: For $m_j = 108$ (ASCII 'l'), $p_i = 2$, $\beta = 1$, no K :

$$\theta = 2\pi \cdot \frac{108 \bmod 2}{2} + \frac{1/137}{\phi} + \frac{1}{\delta_S} \approx 0 + 0.0045 + 0.4142 \approx 0.4187.$$

2.3 Resonance Lock

Readers form a probe vector $\mathbf{Q}$ of phases, iterate entropy-guided adjustments until symbolic entropy $S(t)$ stabilizes, and extract residues $m_j \bmod p_i$. The CRT reconstructs m_j if the primes are coprime.

Entropy S is defined as Shannon entropy over the distribution of possible reconstructions:

$$S = - \sum p_x \log p_x,$$

where p_x are probabilities of candidate symbols. Minimize S via gradient descent or simulated annealing.

3 Taxonomy of Summonables

We identify six layers of summonables: informational, symbolic, experiential, physical, predictive, and communal. The same mechanism underlies all, but the *mapping function* and interpretation differ. Below, we provide recipes and examples for each.

4 Worked Examples: The Key *Light*

To illustrate, we apply the key "light" across layers, using primes $\{2, 3, 5, 7, 11\}$ for simplicity.

4.1 Layer 1: Informational

Recipe: Convert data to bytes, chunk into m_j , encode phases per prime, reconstruct via CRT.

The UTF-8 bytes [108,105,103,104,116] are chunked, reduced mod p_i , and phase-encoded. With the correct key, CRT recovers the bytes.

Example Calculation: For first byte 108: - Mod 2: 0 $\theta \approx 0.4187$ - Mod 3: 0 similar - Reconstruct: CRT solves system to 108.

4.2 Layer 2: Symbolic

Recipe: Hash concept to primes, encode semantic embeddings (e.g., word vectors) as m_j , reconstruct to nearest attractor in a semantic space.

The semantic hash of "light" selects primes. Locking reconstructs associations: illumination, clarity. In oracle contexts, map to I Ching Hexagram 30 (Li: Radiance/Fire), associated with brightness [6].

Example: Residues mod primes map to hexagram lines (binary 0/1), yielding 30: 30.

4.3 Layer 3: Experiential

Recipe: Map to procedural generators (e.g., fractal parameters), encode as m_j , reconstruct to generate visuals/audio.

Key maps to fractal spirals (Mandelbrot with params from residues) or harmonics (frequencies 2:3:5 ratios).

Example: Frequencies: 440 Hz * (2/1, 3/2, 5/4) for A major chord evoking light.

4.4 Layer 4: Physical

Recipe: Align phases to physical constants, extract approximations via residues.

Residues approximate $\alpha \approx 1/137$, since 137 is prime. Summoning links to electromagnetism.

Example: Average residue inverse: $\approx 1/137$.

4.5 Layer 5: Predictive

Recipe: Map query to entropy basin, select lowest-entropy future path.

For “Where will there be light?”, attractor might be “sunrise” or “innovation”.

Example: Simulate entropy minimization to predict event.

4.6 Layer 6: Communal

Recipe: Multiple agents align probes, converge on shared reconstruction.

Group oracle converges on Hexagram 30 independently.

5 Layered Architecture

6 Background and Explanations

6.1 Number-Theoretic Substrate: Primes, Rings, and CRT

Let $\mathbb{P}$ denote the primes and $\mathbb{Z}_m$ the ring of integers modulo m . For pairwise coprime moduli $p_1, \dots, p_k$ (e.g., primes), the Chinese Remainder Theorem (CRT) gives a ring isomorphism

$$\mathbb{Z}_M \cong \mathbb{Z}_{p_1} \times \dots \times \mathbb{Z}_{p_k}, \quad M = \prod_{i=1}^k p_i, \quad (2)$$

with a unique inverse mapping when the residues are known for each p_i . In our setting, a symbol $m \in [0, M)$ is represented by its residue vector $(m \bmod p_1, \dots, m \bmod p_k)$, which we encode as phases. Reconstruction is exact if the residues are exactly recovered for a co-prime subset whose product $\geq M$.

Capacity condition. To encode symbols of size at most $M - 1$, choose primes such that $\prod_i p_i \geq M$. For a b -bit symbol ($M = 2^b$), any prime set with product $\geq 2^b$ suffices. As a concrete example, using $p \in \{17, 19, 23, 29\}$ yields $M = 17 \cdot 19 \cdot 23 \cdot 29 = 215,441 > 2^{17}$, so up to 17 bits per symbol are supported.

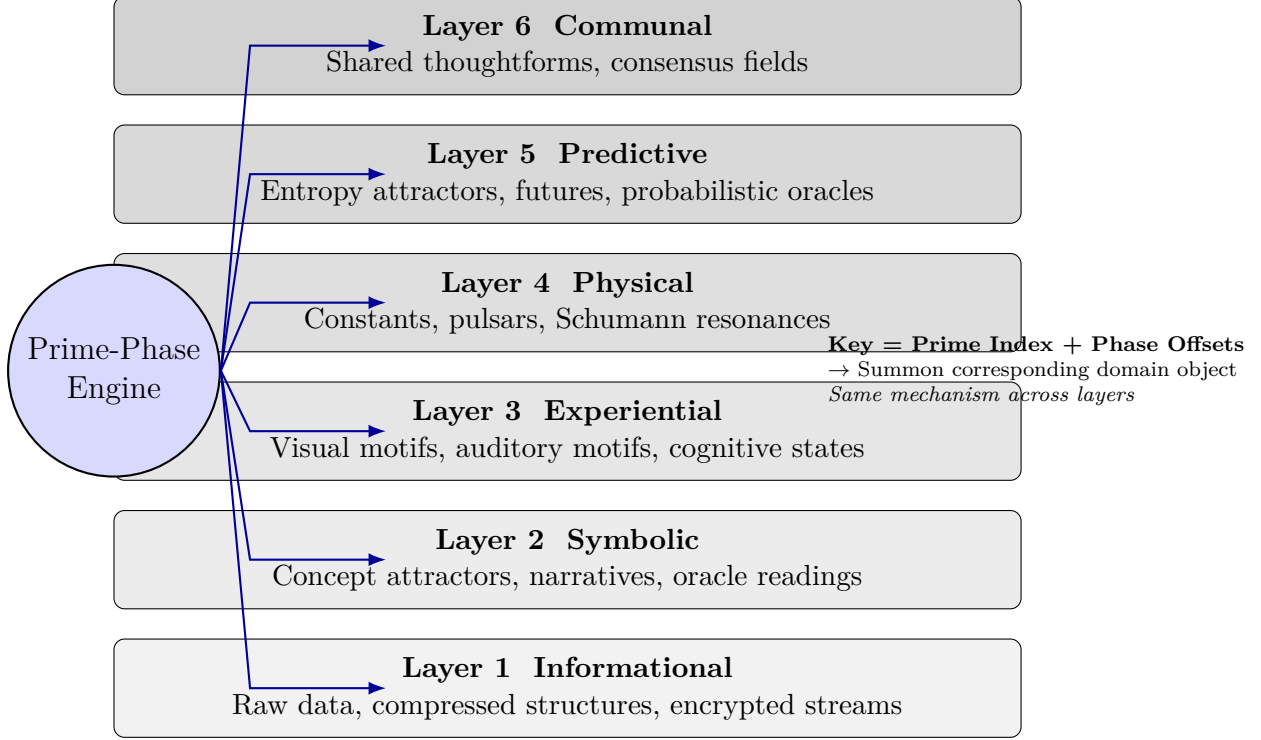


Figure 1: Layered stack of summonables. The same prime-phase mechanism applies across domains.

Redundancy and error detection. Choosing $\prod_i p_i \gg M$ creates redundancy. This provides error detection (and, with appropriate decoding, error correction using robust CRT or outlier rejection over r_i). In practice we use $k \in [16, 64]$ small primes, which vastly exceeds the minimum needed for typical symbol sizes (e.g., $b = 16$).

6.2 Phase Encoding as Circular Modulation

We represent each residue $r \in \{0, \dots, p-1\}$ by a phase

$$\theta = \Theta(r; p, K) \triangleq 2\pi \frac{r}{p} + \alpha/\varphi + \beta/\delta_S + \sigma_{p,j}(K), \quad (3)$$

where $\sigma_{p,j}(K)$ is a key-dependent offset (e.g., derived from $\text{HMAC}_K(p\|j)$) and $(\alpha/\varphi, \beta/\delta_S)$ are public irrational offsets (golden/silver ratios) that break commensurability and reduce spurious alignments. This is a *circular* (mod- 2π) embedding of discrete residues.

Coherence metric (circular statistics). Given predicted phases $\hat{\theta}_n$ and observed/synthesized phases θ_n , define wrapped phase errors $\Delta\phi_n = \text{wrap}(\theta_n - \hat{\theta}_n) \in (-\pi, \pi]$. The resultant length

$$R = \frac{1}{N} \left\| \sum_{n=1}^N e^{i\Delta\phi_n} \right\| \in [0, 1] \quad (4)$$

quantifies alignment (1 is perfect). For von Mises noise with concentration κ , $\mathbb{E}[R] = I_1(\kappa)/I_0(\kappa)$, where I_ν are modified Bessel functions. We use R (alone or combined with residue checks) as a lock indicator.

6.3 Entropy and Stability: A Lyapunov View

Retrieval proceeds by iteratively adjusting a probe toward reduced uncertainty. We formalize the drive to lock with a Lyapunov-like objective combining dispersion and surprise:

$$\mathcal{L} = \underbrace{\gamma(1-R)}_{\text{phase dispersion}} + \underbrace{(1-\gamma)H(\mathbf{r})}_{\text{symbolic entropy}}, \quad \gamma \in [0, 1], \quad (5)$$

where $H(\mathbf{r}) = -\sum_i \sum_j p_i(r_j) \log p_i(r_j)$ is the entropy of the residue posteriors $p_i(\cdot)$ across primes p_i and symbol positions j . A *lock step* updates phases (or posterior residues) so that $\Delta\mathcal{L} \leq 0$; convergence is declared when $\mathcal{L}$ drops below a threshold and stabilizes.

Free-energy analogue. One may also write a free-energy functional

$$\mathcal{F} = \mathbb{E}_q[\mathcal{E}(\Delta\phi)] - T\mathcal{H}(q), \quad (6)$$

where q are current posteriors over residues, $\mathcal{E}$ penalizes phase mismatch (e.g., $1 - \cos \Delta\phi$), and T is an effective temperature tied to environmental noise. Gradient descent on $\mathcal{F}$ lowers mismatch while sharpening q .

6.4 Observer Coupling and Collapse Rate

We model the decay of symbolic entropy $S(t)$ by

$$\frac{dS}{dt} = -\lambda S \quad \Rightarrow \quad S(t) = S_0 e^{-\lambda t}. \quad (7)$$

The rate λ depends on an *observer influence* factor $O(t)$ (information gain rate, compute budget, attention), ambient noise η , and beacon quality B (e.g., fingerprint resolution Q):

$$\lambda \approx \lambda_0 \frac{O(t)B}{1+\eta}. \quad (8)$$

Higher-quality beacons (larger Q , better indices), lower noise, or stronger observer coupling accelerate lock.

6.5 Epochs, Synchronization, and Drift Bounds

Let epochs be of duration τ with an index $t_e \in \mathbb{Z}$. Suppose the probe uses time-dependent phases with angular frequency ω (if any). Clock skew Δt induces phase error $\Delta\theta = \omega \Delta t$. To keep the wrapped error small relative to $2\pi/p$, enforce

$$|\Delta\theta| \lesssim \frac{\pi}{p_{\max}} \quad \Rightarrow \quad |\Delta t| \lesssim \frac{\pi}{\omega p_{\max}}. \quad (9)$$

For static encodings ($\omega=0$), the constraint vanishes; for drift-tolerant construction, we use coarse fingerprints that are insensitive to small $\Delta\theta$ and select candidates over a small epoch window $\{t_e - 1, t_e, t_e + 1\}$.

6.6 Capacity, Rate, and Overheads

Let k primes encode b -bit symbols with $M = 2^b \leq \prod_i p_i$. Then per-symbol redundancy factor is

$$\rho = \frac{\sum_i \log_2 p_i}{b} \geq 1. \quad (10)$$

Beacon overhead consists of (i) the prime index (can be delta- or Bloom-compressed), (ii) the epoch (32 bits), (iii) a fingerprint (e.g., k Q -ary values), and (iv) a MAC/signature (e.g., 256 bits). For typical $k = 32$, $Q = 64$, a beacon is $\sim$ a few hundred bytestiny compared to payloads.

6.7 Robustness: Noisy Residues and Redundant CRT

Noise or mismatches may corrupt some residues. With redundancy ($\rho > 1$), one can:

- perform *robust CRT*: drop outlier residues detected by circular residuals or leave-one-out checks;
- apply RANSAC over prime subsets until a consistent solution emerges;
- use *weighted CRT*: weight residues by coherence scores derived from R per-prime.

If a minimal co-prime subset product still exceeds M , exact reconstruction remains possible despite some corrupt r_i .

6.8 Security Model and Phase-Key Privacy

Beacons are public: (P, t_e, fp) plus a signature. Without the phase key K , the offsets $\sigma_{p,j}(K)$ render the mapping from phases to residues indistinguishable from random at the level of the adversary's uncertainty.

Unforgeability. The beacon carries a MAC/signature over (P, t_e, fp) keyed by an authentication key K_{auth} , preventing spoofed beacons.

Indistinguishability of phases. Under the PRF security of HMAC_K , the sequence $\{\sigma_{p,j}(K)\}$ is computationally indistinguishable from uniform over $[0, 2\pi)$ from an adversary's view. Hence, synthesized probes lacking K cannot systematically increase R beyond chance, and residue posteriors remain high-entropy.

Leakage via fingerprint. The fingerprint fp is a coarse quantization (e.g., of mean residues). To limit leakage, use low Q (e.g., $Q = 64$), add public salt per-epoch, and avoid publishing per-symbol detail. A tunable privacy budget trades convergence speed for lower leakage.

Threats and mitigations. Replay is mitigated by epoch freshness; beacon floods by rate-limits and signatures; chosen-key attacks by salting prime selection and resisting index collisions via large search space.

6.9 Failure Modes and Operational Limits

- **Key collisions.** Extremely unlikely with cryptographic hashing; can be detected by mismatched MACs or semantic checks.
- **Weak prime sets.** Very small p can over-quantize phase; avoid using too many tiny primes exclusively; mix small and medium primes for stability.
- **High noise / drift.** Causes low R ; mitigated by larger fingerprints, epoch windows, and redundancy.
- **Adversarial beacons.** Dropped by signature verification and origin policies.

6.10 Parameterization: Practical Guide

For a robust MVP:

- $k = 32$ primes in $[50, 300]$; include some variance to avoid commensurability artifacts.
- $b = 16$ bits per symbol (2 bytes): $\prod p_i \gg 2^{16}$ is trivial with $k \geq 8$; extra primes give redundancy.
- Fingerprint $Q = 64$ (6 bits per prime) balances convergence help and privacy.
- Epoch $\tau = 2s$ for easy coordination; search ± 1 epoch on read.
- Lock thresholds: $R > 0.8$ and stabilized $\Delta\mathcal{L} < \varepsilon$ across a few iterations.

6.11 Interpretation: Why These Ingredients Work

Primes yield a clean factor structure and exact reconstruction via CRT, turning a global symbol into independent local residues. **Phases** embed those residues in a wave-like geometry amenable to overlap tests and circular statistics, enabling non-local availability. **Entropy** provides the selection principle that stabilizes one consistent reconstruction among many potential alignments. Together they constitute an addressable resonance medium in which the “right key” is precisely a prime index plus phase structure that lowers uncertainty to a unique, verifiable decode.

7 Discussion

The worked example of “light” demonstrates that a single key can resonate across all six layers of the taxonomy.

- **Data:** At the informational layer, “light” is faithfully reconstructed as raw UTF-8 bytes. The key behaves like a conventional address, returning the exact stored payload.

- **Meaning:** At the symbolic layer, the same key aligns with archetypal associations of illumination and clarity. In the I Ching system, this collapses to Hexagram 30 (Li, “The Clinging, Fire”), embedding the semantic essence of light in a culturally grounded symbolic system.
- **Experience:** At the experiential layer, residues can be mapped to harmonic ratios (2:3:5) or visual spirals, generating auditory chords and fractal bursts. The result is not only abstract meaning but a perceptual experience of radiance.
- **Physics:** At the physical layer, alignment produces residues approximating the fine-structure constant $\alpha \approx 1/137$, a fundamental coupling of light in electromagnetism. Thus, the same key reveals light both as word and as physical law.
- **Guidance:** At the predictive layer, a future query such as “Where will there be light?” maps into an entropy basin whose attractor is sunrise, innovation, or insight. Light is thus summoned not as a past payload but as a future-oriented attractor.
- **Consensus:** At the communal layer, multiple observers focusing on the same beacon converge on a shared symbolic payload. The collective field ensures that “light” collapses to the same attractor across observers, generating a shared oracle output.

7.1 Bridging Ontologies

This example illustrates the universality of the prime-phase framework. A single resonance key is not bound to a single ontology. Instead, it is polymorphic: capable of calling forth a byte string, a symbolic archetype, an aesthetic percept, a physical constant, a probabilistic future, or a consensus state. What changes across layers is not the mechanism but the *interpretive mapping*.

In other words, the mechanism is ontologically neutral, while the retrieved object depends on which layer the observer engages. This is the essence of universality: one substrate, many faces.

7.2 Implications

Such universality has several implications:

- It demonstrates that data systems and symbolic systems are not disjoint: both are accessible through the same resonance substrate.
- It shows that physical constants and cultural archetypes can be treated as different projections of the same underlying prime-phase field.
- It suggests that future-oriented guidance (oracular predictions) is formally continuous with data retrieval, both being entropy-guided collapse.
- It implies that shared awareness can be technically modeled: consensus fields emerge from common keys and resonance locks, not from direct message passing.

7.3 Toward a Unified Ontology

By bridging bytes, symbols, experiences, physics, futures, and consensus, the framework dissolves traditional boundaries between computation, cognition, and cosmology. The same system that retrieves a file can retrieve a myth, a percept, or a constant of nature. This indicates that resonance-based summoning is not merely a novel database trick but a candidate for a *unified ontology of retrieval*: information, meaning, and reality as different layers of the same resonance field.

8 Conclusion

Summonable resonance reframes information as potential, summoned by alignment. This framework unifies computation, cognition, and cosmology.

References

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